

React Hooks Assignment

Task 1: useState()

Task: Create a **Student Name App**.

Requirements:

- Create an input field for the student's name.
- Store the input value using `useState()`.
- Display the entered name below the input.
- Add a button to clear the name.

Task 2: useEffect()

Task: Create a **Digital Clock**.

Requirements:

- Display the current time.
- Use `useEffect()` and `setInterval()`.
- Update the time every second.
- Display the time in HH:MM:SS format.
- Clear the interval when the component is removed.

Task 3: useContext()

Task: Create a **Student Information App**.

Requirements:

- Create a `StudentContext`.
- Store the following information in the context:

None

Name

Course

Batch

Sample data:

None

Name: Kamal

Course: Full Stack Development

Batch: Cohort 2

- Create three separate components: App, Home, and Student.
- Use `useContext()` to display the student information in the Student component.
- Do **not** pass the student information using props.

Task 4: useRef()

Task: Create an **Input Focus App**.

Requirements:

- Create a text input.
- Create a `useRef()` reference for the input.
- Add a **Focus** button that automatically focuses the input when clicked.
- Add a **Clear** button that clears the input.

Task 5: useReducer()

Task: Create a **Shopping Cart Quantity App**.

Requirements:

- Display the product name, product price, and quantity.
- Add an **Increase Quantity** button.
- Add a **Decrease Quantity** button.
- Add a **Reset** button.
- Calculate and display the total price.
- Use `dispatch()` to handle the different actions.

Expected output format:

None

Product: Notebook

Price: Rs. 500

Quantity: 2
Total: Rs. 1000

Task 6: useMemo()

Task: Create a **Product Price Calculator**.

Requirements:

- Create inputs for product price, quantity, and discount.
- Calculate the final price using `useMemo()`.
- The result must update when the price, quantity, or discount changes.

Formula:

None

$\text{Total} = \text{Price} \times \text{Quantity}$

$\text{Discount Amount} = \text{Total} \times \text{Discount} / 100$

$\text{Final Price} = \text{Total} - \text{Discount Amount}$

Task 7: useCallback()

Task: Create a **Student List App**.

Requirements:

- Create a list of students.
- Display the students using `map()`.
- Create a separate Student component.
- Create a function to handle selecting a student.
- Use `useCallback()` for that function.
- When a student is clicked, display:

None

Selected Student: Kamal

